Claudia Rella

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Institut des Hautes Études Scientifiques (IHES)
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RESEARCH PROFILE

I am a mathematical physicist working at the interface of quantum field theory, string theory, and mathematics. My research seeks to uncover the hidden structures that govern quantum theories beyond perturbation theory, drawing on ideas from algebraic geometry, number theory, topology, and homological algebra. In recent years, I have been developing the programme of modular resurgence, a new research framework connecting resurgence, arithmetic, and quantum modularity and revealing unexpected exact symmetries in quantum physics and quantum topology.

EMPLOYMENT

2024 – present **Postdoctoral Researcher in Mathematical Physics**
Institut des Hautes Études Scientifiques (IHES), France
Mentored by **Prof. Maxim Kontsevich** and funded by the **Huawei Young Talents Program**

EDUCATION

2020 – 2024 **PhD in Mathematical Physics** — Très Bien (highest honours)
Department of Theoretical Physics, **Université de Genève**, Switzerland
Thesis: *On the arithmetic of resurgent topological strings*
Supervised by **Prof. Marcos Mariño** and funded by the **ERC SyG ReNewQuantum**

2018 – 2019 **MSc in Mathematical and Theoretical Physics** — Distinction (highest honours)
Mathematical Institute and Department of Physics, **University of Oxford**, UK
Thesis: *An introduction to motivic amplitudes*
Supervised by **Prof. Francis Brown** and funded by the **Torno Subito Scholarship**

2015 – 2018 **BSc in Physics** — Summa cum Laude (highest honours)
Department of Physics, **Sapienza Università di Roma**, Italy

PUBLICATIONS

PREPRINTS

[2] *Modular resurgent structures for vectors,* with M. C. N. Cheng, I. Coman, and V. Fantini [arXiv: 2608.08902]

[1] *Modular resurgent structures,* with V. Fantini, submitted [arXiv:2404.11550]

REFEREED RESEARCH ARTICLES

[7] *Modular resurgence, q-Pochhammer symbols, and quantum operators from mirror curves*, with V. Fantini, **Lett. Math. Phys.** 116, 39 (2026), https://doi.org/10.1007/s11005-026-02063-x [arXiv:2506.08265]

[6] *On the structure of wave functions in complex Chern–Simons theory*, with M. Mariño, **SIGMA** 22 (2026), 002, 45 pages, https://doi.org/10.3842/SIGMA.2026.002 [arXiv:2312.00624]

[5] *Strong-weak symmetry and quantum modularity of resurgent topological strings on local $\mathbb{P}^2$*, with V. Fantini, **Commun. Num. Theor. Phys.** 19 (2025), no. 1, pp. 1-73, https://dx.doi.org/10.4310/CNTP.250215002033 [arXiv:2404.10695]

[4] *Resurgence, Stokes constants, and arithmetic functions in topological string theory*, **Commun. Num. Theor. Phys.** 17 (2023), no. 3, pp. 709-820**,** https://dx.doi.org/10.4310/CNTP.2023.v17.n3.a4 [arXiv:2212.10606]

[3] *Searching for muonphilic dark sectors with proton beams*, with B. Döbrich and T.-T. Yu, **Phys. Rev. D** 106 (2022) 3, 035023, https://doi.org/10.1103/PhysRevD.106.035023 [arXiv:2205.09870]

[2] *An introduction to motivic Feynman integrals*, **SIGMA** 17 (2021), 032, 56 pages, https://doi.org/10.3842/SIGMA.2021.032 [arXiv:2009.00426]

[1] *Characterization and performance of PADME's Cherenkov-based small-angle calorimeter*, with A. Frankenthal et al., **Nucl. Instrum. Meth. A** 919 (2019) 89-97, https://doi.org/10.1016/j.nima.2018.12.035 [arXiv:1809.10840]

PROCEEDINGS

Mini-Workshop: Resurgence, Difference Equations and Quantum Modularity, with M. Alim, V. Fantini, and L. Hollands, **Oberwolfach Rep.** 22 (2025), no. 4, pp. 2593-2632 (main reporter), https://doi.org/10.4171/owr/2025/48

DOCTORAL THESIS

On the arithmetic of resurgent topological strings, Doctoral Thesis Sc. 5852, **Université de Genève** (2024), https://dx.doi.org/10.13097/archive-ouverte/unige:181979

INVITED TALKS AND SEMINARS

2027, Jan	Conference on Resurgence Theory in Mathematical Physics, **TSIMF**, Sanya, China (forthcoming)
2026, Nov	Mathematical Physics Seminar, **LPTHE**, France (forthcoming)
Oct	11th Workshop on Cosmology and High-Energy Physics, **Université de Montpellier**, France
Mar	Workshop on Mathematical Structures of Amplitudes, **ESI**, Austria
2025, Nov	Asymptotics, Geometry, and Physics Seminar, **University of Southampton**, UK
Oct	Mini-Workshop on Resurgence, Difference Equations, and Quantum Modularity, **MFO**, Germany
Aug	ReNewQuantum Closing Conference, **SDU**, Denmark
May	Random Matrix Theory Seminar, **University of Oxford**, UK
May	Mathematical Physics and Algebraic Geometry Seminar, **CMSA**, **Harvard University**, USA
Apr	Conference on Higher Structures, Moduli Spaces and Integrability, **Universität Hamburg**, Germany
Mar	Workshop on The Arithmetic of Calabi-Yau Manifolds, **MITP**, Germany
Jan	Youngst@rs Physics and Number Theory, **MITP**, Germany
2024, Nov	The Seed Seminar of Mathematics and Physics, **IHP**, France
Oct	Workshop on Holonomic Techniques for Feynman Integrals, **MPI Physics**, Germany
Sep	GAP XIX: Moduli Spaces and Higher Structures, **Sapienza Università di Roma**, Italy
May	String Theory Seminar, **Université de Genève**, Switzerland
May	Workshop on Resurgence and Modularity in QFT and String Theory, **GGI**, Italy
May	Mathematics Seminar, **Yale University**, USA
Apr	String Theory Seminar, **CERN**, Switzerland
Mar	Seminar on Quantum Modularity and Resurgence, **IHES**, France
Feb	Workshop on Positive Geometry in Particle Physics and Cosmology, **MPI MiS**, Germany
2023, Dec	Geometry and Physics Seminar, **University of Sheffield**, UK
Nov	QM Research Seminar, Centre for Quantum Mathematics, **SDU**, Denmark
Jul	29th Nordic Congress of Mathematicians with **EMS**, Aalborg, Denmark
Apr	Mathematics Seminar, **IHES**, France
Mar	Physical Mathematics Seminar, **Université de Genève**, Switzerland
2022, Nov	Workshop on Mathematics of Beyond All-Orders Phenomena, **University of Cambridge**, UK
2019, Nov	Seminar on Lie Groups and Moduli Spaces, **Université de Genève**, Switzerland
Aug	Conference on Representation Theory and Integrable Systems, **ETH Zürich**, Switzerland
2017, Dec	PADME Weekly Seminar, **INFN – LNF**, Italy

TEACHING EXPERIENCE

2027, Jul — Lectures on Asymptotic Series and Resurgence for doctoral students (forthcoming)
Amplitudes School, Johannes Gutenberg Universität Mainz, Germany

2024, Jan — Exercise sessions on Feynman Integrals and Number Theory for doctoral students
Winter School in Mathematical Physics, SwissMAP Research Station, Switzerland

2019, Oct — Lectures on Topological Surfaces for master's students
Master Class in Mathematical Physics, Université de Genève, Switzerland

2018, Mar – May — Lectures on Riemannian Geometry for bachelor's students
Excellence Program in Physics, Sapienza Università di Roma, Italy

HONOURS, FELLOWSHIPS, AND AWARDS

2026 — L'Oréal–UNESCO For Women in Science Young Talent Award, Académie des Sciences, France
2024 – 2027 — Huawei Young Talents Program, IHES, France
2019 – 2020 — Excellence Fellowship, NCCR SwissMAP, Switzerland
2019 — Degree Prize for Distinction, St John's College, University of Oxford, UK
2018 – 2019 — Torno Subito Scholarship, Regione Lazio, Italy
2018 — Best Student in Nuclear and Subnuclear Physics, Sapienza Università di Roma and INFN, Italy
2017 — Summer Student Scholarship, INFN, Italy
2016 – 2018 — Excellence Program, Department of Physics, Sapienza Università di Roma, Italy
2015 – 2018 — Merit-Based Full Tuition Waiver, Sapienza Università di Roma, Italy

SCIENTIFIC ACTIVITIES

COMMUNITY ENGAGEMENT

2026 – present — Co-organiser of the monthly Lunch for Women Scientists at IHES, France

PROFESSIONAL SERVICE

2026 — Co-organiser of a one-day workshop at Laboratoire de Mathématiques d'Orsay, France (forthcoming)
2026 – present — Referee for Communications in Number Theory and Physics
2025 — Co-organiser of a five-day workshop at Mathematisches Forschungsinstitut Oberwolfach, Germany
2020 – 2024 — Junior Member of the Scientific Council of the SwissMAP Research Station, Switzerland

RESEARCH NETWORKS

2024 – present — Associated Researcher, ERC SyG UNIVERSE+
2020 – 2024 — Doctoral Student, ERC SyG ReNewQuantum
2019 – 2024 — Excellence Fellow and Doctoral Student, NCCR SwissMAP

RESEARCH INTERNSHIPS

2020 — Research Internship in High-Energy Physics Phenomenology
CERN, Switzerland

2019 – 2020 — Master Class in Mathematical Physics
NCCR SwissMAP and Université de Genève, Switzerland

2017 — Research Internship in Experimental Particle Physics
Laboratori Nazionali di Frascati, INFN, Italy

SKILLS

Languages — Italian (native), English, French
Programming — C/C++, Python, R
Symbolic Computation — Mathematica, MATLAB, SageMath, PARI/GP